BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 235 · Software development: practical testing

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a prototype?

Q2. Why might a tester run tests at each sprint?

Q3. What does AND require for a compound condition to be true?

Q4. For valid ages 12–16 inclusive, which four values test either side of both boundaries?

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I Do · Read new knowledge

Key vocabulary

normal test — Typical valid input.

boundary test — Input at or near a valid limit.

erroneous test — Input outside permitted requirements.

expected result — Predicted correct output for a test.

regression — New change breaks previously working behaviour.

New learning

State the valid range — The specification permits whole-number marks from 0 through 100 inclusive.

Choose targeted tests — Normal value 50; valid boundaries 0 and 100; invalid just beyond boundaries −1 and 101.

Write expected results — A useful test log records input, expected outcome and actual outcome; a mismatch is evidence of a fault.

Worked example

Design tests for a password length from 8 through 20 characters inclusive.

1. Predict first — Test just below and at the lower limit: 7 rejected, 8 accepted.

2. Model and explain — Also test a normal 14, upper boundary 20, and 21 rejected.

We Do · Practise with guidance

Choose boundary and invalid tests for a class size of 1–30 inclusive.

```
static bool ValidSize(int n)
    => n >= 1 && n <= 30;
```

Predict first

Lower boundary 1 is accepted while 0 is invalid.

Model and explain

Upper boundary 30 is accepted while 31 is invalid; 16 is a representative normal value.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

A whole-number temperature must be 10–40 inclusive. Give a normal, a lower-boundary, an upper-boundary and an erroneous test.

Question 2

Write C# to validate a score from 1 to 9 inclusive and give four expected-result test pairs.

Question 3

A developer tests values 20,30,40 for a permitted range 20–40. Explain why this is insufficient and propose a stronger suite.

Exit Ticket · Answer before checking

Testing only typical valid inputs proves that validation rejects out-of-range values.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. An early model that users evaluate and refine.

Q2. To catch regressions while new features are introduced.

Q3. Both conditions must be true.

Q4. 11, 12, 16 and 17.

You Do pages · Independent questions

Q1. [4] Example 25 [1]; 10 [1]; 40 [1]; 9 (or 41) [1].

Q2. [6] bool Valid(int x)=>x>=1&&x<=9; [2]. Test 0 false [1], 1 true [1], 9 true [1], 10 false [1].

Q3. [6] Existing tests include boundaries [2] but no immediately invalid values [1]; add 19 and 41 expected rejection [2], and a representative interior value such as 27 expected acceptance [1].

Exit Ticket · Model answer

False. Erroneous and boundary-adjacent tests are also needed.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: